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Project Report

Project Title: Retinal Blood Vessel Segmentation using Principal Curvature Method in MATLAB

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ABSTRACT

Retinal blood vessel segmentation is a critical task in the field of ophthalmology, aiding in the early detection and monitoring of various ocular diseases such as diabetic retinopathy and glaucoma. Accurate segmentation of retinal blood vessels is challenging due to the presence of complex vessel structures, variations in vessel width, intensity variations, and the presence of pathologies. Various image processing techniques, machine learning algorithms, and deep learning approaches have been employed to tackle this problem. Our project focuses on the utilization of the principal curvature approach for retinal blood vessel segmentation using MATLAB.

The principal curvature method leverages the concept of vessel tubularity to distinguish blood vessels from the background in retinal images. By computing the principal curvature on the retinal image, regions with high curvature values are identified as potential vessel segments. These segments are then further refined using additional processing steps, such as thresholding and morphological operations, to obtain accurate vessel boundaries.

The implementation of the principal curvature-based approach in MATLAB provides a practical and efficient solution for retinal blood vessel segmentation. MATLAB's rich set of image processing functions and tools enable effective computation of principle curvature and subsequent vessel extraction. Moreover, MATLAB's versatility allows for easy integration with other image processing techniques, further enhancing the accuracy and robustness of the segmentation results.

The principal curvature-based approach shows promise in improving the segmentation of retinal blood vessels, which in turn contributes to enhanced diagnostic capabilities and more precise monitoring of ocular diseases. This technique has the potential to assist ophthalmologists in early disease detection, treatment planning, and follow-up assessments.

List of Abbreviations & Symbols

MATLAB	Matrix Laboratory
CLAHE	Contrast-limited adaptive histogram equalization
TPR	True Positive Rate
FPR	False Positive Rate
TP	True Positive
TN	True Negative
FP	False Positive
FN	False Negative
OCT	Optical Coherence Tomography

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Chapter 1

Introduction & Objectives

1.1 Introduction

1.1.1 Retinal Vascular System

The retinal vascular system provides rich information about the state of the eye and is the only non-invasive imaging method to obtain visible blood vessels from the human body. Retinal vascular segmentation is of great significance for the diagnosis of fundus diseases. As a result, retinal images have been widely used to detect early signs of systemic vascular disease. In order to facilitate the diagnosis of systemic vascular diseases, vessels need to be accurately segmented. Therefore, the automatic segmentation of retinal blood vessels from fundus images has become a popular research topic in the medical imaging field.

In this project, retinal blood vessels from fundus images are used for retinal blood vessel segmentation. The implementation of image processing techniques such as contrast enhancement and area opening are used for retinal blood vessel segmentation. This project uses principal curvature method for segmentation. This method measures the principal curvature of the image intensity surface and generates a single response for both lines and edges, producing a clearer structural sketch of an image than is usually provided by the gradient magnitude image. The main aim of this project is to perform segmentation of retinal blood vessels using MATLAB.

1.1.2 Segmentation

Segmentation partitions an image into its constituent parts or objects. In general, autonomous segmentation is one of the most difficult tasks in digital image processing. A rugged segmentation procedure brings the process a long way toward successful solution of imaging problems that require objects to be identified individually. On the other hand, weak or erratic segmentation algorithms almost always guarantee eventual failure. In general, the more accurate the segmentation, the more likely automated object classification is to succeed.

1.1.3 Ground Truth

Ground truth is a term used in statistics and machine learning that means checking the results of machine learning for accuracy against the real world. The term is borrowed from

meteorology, where “ground truth” refers to information obtained on site. The term implies a kind of reality check for machine learning algorithms.

1.1.4 Image Masking

Image masking is actually a photo editing process that enables you to eliminate the background of an image. It is a non-destructive way to replace the original background and make the image unique. Typically, image masking is the process of limiting a certain visible adjustment in photos.

With this technique, you can alter a particular portion of an image without editing the entire background. For instance, you want to brighten an object of your image while leaving the background unedited. In that case, image masking lets you precisely control and mask your desired object to brighten it.

1.1.5 Erosion

Erosion is a morphological operation that is used to shrink the objects in an image. It is done by repeatedly moving a structuring element over the image and setting the value of a pixel to 0 if any of its neighbors are 0. Erosion can be used to remove noise from an image, to extract objects from an image, or to clean up an image.

The structuring element would be a small square or rectangle, and it would be moved over the image one pixel at a time. If any of the pixels in the structuring element were 0, then the corresponding pixel in the image would be set to 0. This would remove any small, isolated pixels from the image, leaving only the larger, more significant pixels.

Erosion is a powerful tool that can be used to improve the quality of images in a variety of ways. It is a relatively simple operation to implement, and it is very effective at removing noise and other unwanted objects from images.

1.1.6 Gaussian Filtering

In digital image processing, a Gaussian filter is a low-pass filter often used to remove noise and smooth out images. It is a linear filter, which means that the output of the filter is a linear combination of the inputs. The Gaussian filter is named after Carl Friedrich Gauss, who developed the Gaussian distribution, which is the probability distribution that is used to create the filter kernel.

The Gaussian filter is a 2-D filter, which means that it operates on images that are represented as a grid of pixels. The filter kernel is a square array of numbers, and the size of the kernel determines the amount of smoothing that is applied to the image. The larger the kernel, the more smoothing is applied.

The Gaussian filter works by convolving the filter kernel with the image. Convolution is a mathematical operation that takes two functions and produces a third function. In the case of the Gaussian filter, the first function is the filter kernel and the second function is the image. The output of the convolution operation is a new image that has been smoothed by the Gaussian filter.

The Gaussian filter is a very versatile filter, and it can be used for a variety of tasks in digital image processing. Some of the most common uses of the Gaussian filter include:

1. Noise removal: The Gaussian filter can be used to remove noise from images. This can be useful for improving the quality of images that have been affected by noise, such as images that have been taken in low-light conditions.
2. Image smoothing: The Gaussian filter can be used to smooth out images. This can be useful for reducing the amount of detail in an image, or for making an image look more natural.
3. Edge detection: The Gaussian filter can be used to detect edges in images. This can be useful for identifying objects in an image, or for segmenting an image into different regions.

The Gaussian filter is a powerful tool that can be used for a variety of tasks in digital image processing. It is a versatile filter that can be used to improve the quality of images, or to extract information from images.

1.1.7 Principal Curvature

Principal curvature is a measure of the local curvature of a surface at a given point. In digital image processing, it can be used to identify edges, corners, and other features in an image. The principal curvatures of a surface can be computed using the Hessian matrix, which is a square matrix of second-order partial derivatives. The eigenvalues of the Hessian matrix are the principal curvatures, and the eigenvectors of the Hessian matrix are the directions of the principal curvatures.

Principal curvature is a powerful tool for image processing, and it has been used in a variety of applications, such as edge detection, corner detection, and image segmentation.

1.1.8 Contrast Enhancement

Contrast enhancement is a technique used to improve the contrast of an image. This can be done by adjusting the brightness and/or saturation of the image. Contrast enhancement can be used to make images more visually appealing, or to improve the visibility of details in an image.

There are a number of different contrast enhancement techniques available. Some of the most common techniques include:

1. Histogram equalization: This technique equalizes the distribution of pixel intensities in an image. This can be done by stretching the histogram of the image so that it covers the full range of possible pixel values.
2. Contrast stretching: This technique stretches the contrast of an image by expanding the range of pixel values. This can be done by setting a minimum and maximum value for the pixel values, and then remapping all of the pixel values to fall within this range.
3. Local contrast enhancement: This technique enhances the contrast of an image locally, by adjusting the brightness and/or saturation of each pixel based on its neighbors. This can be done by using a variety of different algorithms.

Contrast enhancement is a powerful tool that can be used to improve the quality of images. It is a versatile technique that can be used for a variety of different purposes.

1.1.9 Morphological Area Opening

Morphological area opening is a morphological operation that is used to remove small objects from an image. It is performed by first eroding the image, and then dilating the eroded image. Erosion removes pixels from the boundary of objects, while dilation adds pixels to the boundary of objects. The combination of erosion and dilation removes small objects from the image while preserving larger objects.

Morphological area opening is often used to remove noise from images. It can also be used to extract objects from images. For example, if you want to extract all of the white pixels in an image, you could perform morphological area opening using a structuring element that is a single white pixel.

Morphological area opening is a powerful tool that can be used to improve the quality of images. It is a versatile operation that can be used for a variety of different purposes.

Here are some additional details about morphological area opening:

- (i) Structuring element: The structuring element is a small shape that is used to erode and dilate the image. The size and shape of the structuring element determines the size and shape of the objects that will be removed from the image.
- (ii) Erosion: Erosion removes pixels from the boundary of objects. The pixels that are removed are those that are not covered by the structuring element.
- (iii) Dilation: Dilation adds pixels to the boundary of objects. The pixels that are added are those that are covered by the structuring element.

In mathematical morphology, opening is the dilation of the erosion of a set A by a structuring element B :

$$A \circ B = (A \ominus B) \oplus B,$$

Where, $\ominus$ and $\oplus$ denote erosion and dilation, respectively. Morphological area opening is a simple and effective operation that can be used to remove small objects from images. It is a versatile operation that can be used for a variety of different purposes.

1.1.10 Validation check using Ground Truth Image

Ground truth image is a reference image that is used to validate the results of digital image processing algorithms. It is created by manually labeling each pixel in the image with its corresponding ground truth value. The ground truth image is then compared to the output of the algorithm to assess its accuracy. This process is known as validation check.

Validation check is an important step in the development of digital image processing algorithms. It helps to ensure that the algorithm is working as intended and that it is producing accurate results. Validation check can also be used to compare different algorithms and to identify the best algorithm for a particular application. Here are some of the benefits of using ground truth image for validation check:

1. It provides a reference image that can be used to assess the accuracy of the algorithm.
2. It helps to identify errors in the algorithm.
3. It can be used to compare different algorithms.
4. It can be used to identify the best algorithm for a particular application.

1.2 Objectives:

The main objectives of this project are to –

1. To learn about different image processing techniques
2. To segment retinal blood vessels from fundus image using principal curvature method in MATLAB
3. To check the validity of our obtained resulting image using ground image

Chapter 2

Background Study

2.1 Background Study

Retinal blood vessel segmentation plays a crucial role in the diagnosis and management of various retinal diseases, such as diabetic retinopathy, hypertensive retinopathy, and age-related macular degeneration. By accurately identifying and delineating retinal blood vessels, clinicians can gain valuable insights into the vascular health of the eye, assess the progression of diseases, and provide timely intervention. This background study aims to provide an overview of retinal blood vessel segmentation techniques, highlighting their significance and recent advancements.

Retinal blood vessels serve as a vital component of the ocular system, providing essential nutrients and oxygen to the retina. They also serve as an indicator of systemic diseases, such as hypertension and diabetes, which often manifest in the form of retinal vascular changes. By segmenting retinal blood vessels from fundus images, clinicians can quantitatively analyze vessel morphology, measure vessel caliber, detect abnormalities, and assist in disease classification and monitoring.

Early retinal blood vessel segmentation methods relied on handcrafted features and classical machine learning algorithms. These methods typically involved pre-processing steps, followed by feature extraction and classification. Techniques like matched filtering, region growing, and thresholding were employed to extract vessel features based on pixel intensities, morphological characteristics, or vessel connectivity. However, these approaches had limitations in handling complex vessel structures, variability across images, and robustness in the presence of noise.

The advent of deep learning revolutionized the field of retinal blood vessel segmentation. Convolutional neural networks (CNNs) emerged as powerful tools for automatic feature learning and achieved state-of-the-art performance. Fully convolutional networks (FCNs), U-Net, and variations of encoder-decoder architectures became popular for their ability to capture contextual information and spatial dependencies.

Retinal blood vessel segmentation often requires large amounts of annotated data for training. Datasets like DRIVE, STARE, and CHASE_DB1 have been widely used to benchmark segmentation algorithms. However, due to the limitations of manual annotations and the cost of obtaining ground truth, researchers have explored alternative approaches. Techniques such

as transfer learning, semi-supervised learning, and generative adversarial networks (GANs) have been proposed to alleviate the burden of data annotation and improve model generalization.

Researchers continue to explore advanced techniques to enhance retinal blood vessel segmentation. Multi-scale analysis, incorporating contextual information, vessel tracking, and vessel junction detection are among the ongoing efforts to improve the accuracy and robustness of segmentation algorithms. Additionally, the fusion of multimodal imaging data, such as combining fundus images with optical coherence tomography (OCT) or fluorescein angiography (FA), has shown promise in achieving more comprehensive vascular assessments.

2.2 Motivation

Despite significant progress, retinal blood vessel segmentation still faces several challenges. Variations in vessel appearance, vessel tortuosity, presence of pathologies, and image quality issues pose difficulties for accurate segmentation. Addressing these challenges requires the development of robust algorithms, comprehensive datasets, and standardized evaluation metrics. Furthermore, the integration of artificial intelligence-based segmentation tools into clinical practice raises ethical considerations, interpretability, and regulatory challenges. To solve this problem, we used the principal curvature method in this project to segment the blood vessels from retinal image.

Chapter 3

Methodology

3.1 Flowchart:

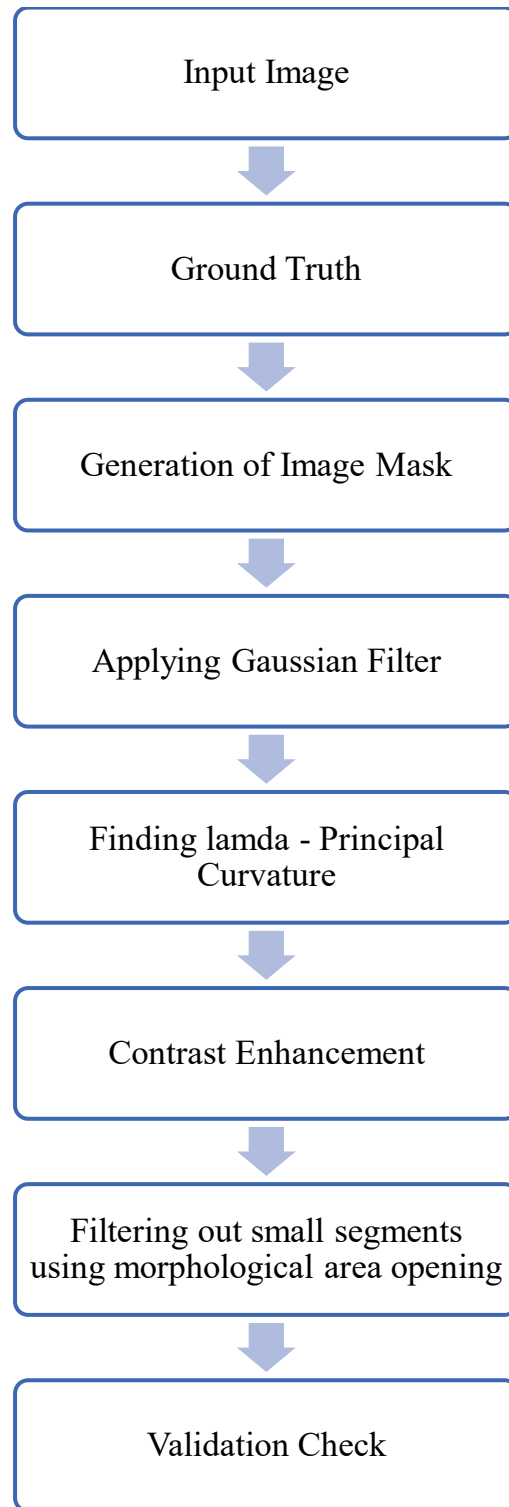


Figure 3.1: The Block diagram of Retinal Blood Vessel Segmentation Procedure

3.2 Procedure Description

According to this basic flowchart, our method will firstly read the image from data base that can be used as input image. It will also read the corresponding ground image. Then a mask will be generated. Using the gaussian filter, we will make smooth and noise free image that is fetched from dataset. Then we will apply the principal curvature method and obtain parameters for Hessian matrix in every pixel and find eigenvalues of the hessian matrix using 'lamdafind' function. After that, Contrast-limited adaptive histogram equalization (CLAHE) will be applied for image contrast enhancement. Small segments will be filtered out using morphological area opening. Then we will get out processed segmented image. At last, we will check the validity of segmentation using the preloaded ground truth image.

2.3 Software required

1. MATLAB R2020a

Chapter 4

Results and Discussions

4.1 Results

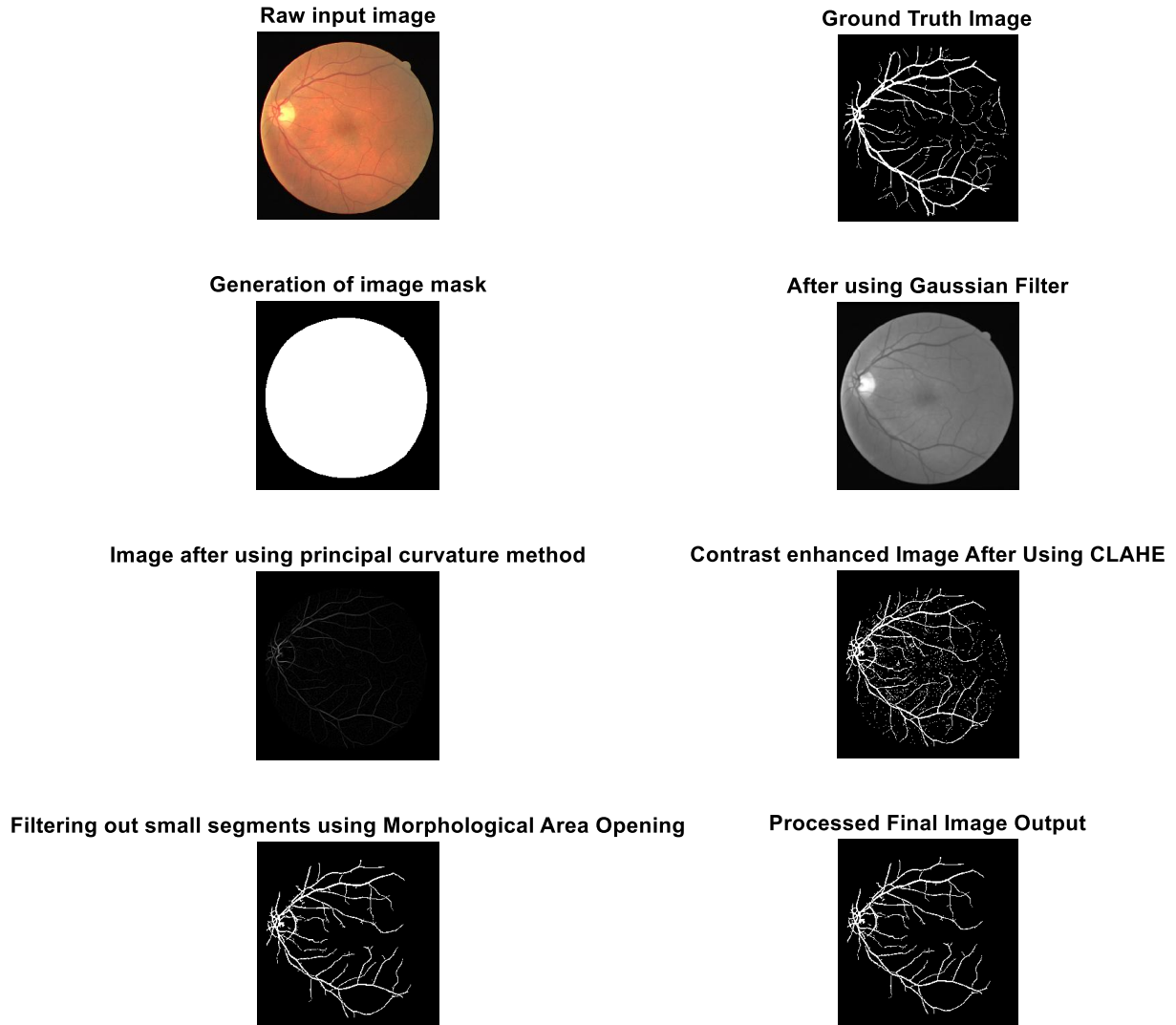


Figure 4.1: Output images of the MATLAB code for Retinal Blood Vessel Segmentation using principal curvature method

Table 4.1: The results of Segmentation using Principal curvature model:

Model	TPR	FPR	Accuracy	Sensitivity	Specificity
{'Principal curvature'}	0.61887	0.010652	0.96166	0.61887	0.98935

4.2 Discussions

In this project, the images were read using 'imread()' function. Gaussian Filter was used using 'imgaussfilt()' function. We used iteration to find the eigen values of hessian matrix. Contrast enhancement using 'adapthisteq()' for Contrast-limited adaptive histogram equalization (CLAHE). Small segments were filtered out using Morphological Area Opening with 'bwareaopen()' function. Then the validity of resulting segmented image was checked using the preloaded ground truth image and the validation parameters, such as, accuracy, sensitivity, specificity, TPR, FPR was calculated. The accuracy for retinal blood vessel segmentation is 96.17%. The true positive rate is 61.89% and false positive rate is 1.07%. The sensitivity and specificity of the segmentation is 61.89% and 98.94% respectively.

Chapter 5

Conclusion and Future work

5.1 Conclusion

This project will assist the healthcare practitioners to detect and monitor various ocular diseases such as diabetic retinopathy and glaucoma at early stage. Here, different image processing techniques were learnt. Retinal blood vessels were segmented from fundus image using principal curvature method in MATLAB. The validity of our obtained resulting image was checked using ground image. Thus, all the objectives of this project were fulfilled.

5.2 Future work

The current algorithm is able to segment the retinal blood vessels with a high degree of accuracy, but there is still room for improvements. So, I want to improve the accuracy of the segmentation algorithm by using a more sophisticated model of the retinal blood vessels in future. I also need to make the algorithm more robust to noise. The segmentation algorithm can be made more efficient by using a GPU to accelerate the computation. The current algorithm is designed to segment retinal blood vessels in fundus images. However, the algorithm can be extended to segment blood vessels in other types of images, such as OCT images and fluorescein angiography images.

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Appendix A: MATLAB Code

Code for segmentation of retinal blood vessels using principal curvature method:

```
clc;
clear;
close all;
inputImage = imread('Data_Set/21_training.tif');
groundTruth = imread('Data_Set/21_manuall.gif');

% Plotting the original input image
subplot 421
imshow(inputImage);
title("Raw input image");

% Plotting the ground truth image
subplot 422
imshow(groundTruth);
title("Ground Truth Image");

%Generation of image mask
mask = im2bw(inputImage,20/255);
%str. element of dimond type of size 20 (errosion)
se = strel('diamond',20);
erodedmask = im2uint8(imerode(mask,se)); %eroded image
subplot 423
imshow(erodedmask);
title("Generation of image mask");

%Apply gaussian filter to the image where s=1.45
img3= imgaussfilt(inputImage(:,:,2) ,1.45);
subplot 424
imshow(img3);
title("After using Gaussian Filter");

%Finding lamda - principal curvature
[gx, gy] = gradient(double(img3));
[gxx, ~] = gradient(gx);
[gxy, gyy] = gradient(gy);
[row,col]=size(img3);
```

```

lamdaplus = zeros(row,col);
lamdaminus = zeros(row,col);
%finding eigen values of hessian matrix [gxx gxy;gxy gyy]
for r = 1:row
    for c = 1:col

[lamdaplus(r,c),lamdaminus(r,c)]=lamdafind(gxx(r,c),gyy(r,c),gxy(r,c));
        end
    end
lamda2 = lamdaplus;
maxprincv = im2uint8(lamda2/max(lamda2(:)));
maxprincvmsk = maxprincv.*(erodedmask/255);
subplot 425
imshow(maxprincvmsk);
title("Image after using principal curvature method");

%Contrast enhancement using Contrast-limited adaptive histogram
equalization (CLAHE)
newprI = adapthisteq(maxprincvmsk,'numTiles',[8 8],'nBins',128);
thresh = isodata(newprI);
vessels = imbinarize(newprI,thresh);
subplot 426
imshow(vessels);
title("Contrast enhanced Image After Using CLAHE");

%Filtering out small segments using Morphological Area Opening
vessels = bwareaopen(vessels, 200);
segImage = vessels;
subplot 427
imshow(segImage);
title("Filtering out small segments using Morphological Area Opening");

subplot 428
imshow(segImage);
title("Processed Final Image Output");

%This function perform validation of vessel segmentation
image=im2uint8(segImage);

%TP - True Positives

```

```

%True positive - Any pixel marked as vessel in both ground truth and
segemented image

%FP - False positive : Any pixel Marked as a vessel pixel in segmented
image
%which is marked as background pixel in ground truth

%TN - True Negative : Any pixel marked as backgroundf in both ground truth
%and segmented image

TP = 0;
TN = 0;
FP = 0;
FN = 0;
vGT = 0;
bGT = 0;

[row,col]= size(groundTruth);

for r=1:row
    for c = 1:col
        if groundTruth(r,c)==255
            vGT=vGT+1;
            if image(r,c)==255
                TP=TP+1;
            end

        else
            bGT=bGT+1;
            if image(r,c)==0
                TN=TN+1;
            else
                FP=FP+1;
            end
        end
    end
end

TPR = TP/vGT; %vGT : Number of vessel pixels - Ground truth
FPR = FP/bGT; %vGT : Number of non-vessel pixels - Ground truth

```

```

FN = (row*col)-(TP + TN + FP);
%Accuracy = (True negative+ True Positive)/all pixels
Accuracy= (TP+TN)/(TP+TN+FP+FN);
True_Positive = TP/(row*col);
True_Negative = TN/(row*col);
False_Positive = FP/(row*col);
False_Negative = FN/(row*col);
Sensitivity = TP/(TP+FN);
Specificity = TN/(TN+FP);
Model = {'Principal curvature'};
A = table(Model, TPR, FPR, Accuracy, Sensitivity, Specificity);
disp(A)

```

Function to compute global image threshold using iterative isodata method:

```

function [level,MAT,MBT]=isodata(I)
% ISODATA Compute global image threshold using iterative isodata method.
% LEVEL = ISODATA(I) computes a global threshold (LEVEL) that can be
% used to convert an intensity image to a binary image with IM2BW.

% Converting all N-D arrays into a single column. Convert to uint8 % for
fastest histogram computation.
I = im2uint8(I(:));

% STEP 1: Computing mean intensity of image from histogram, set T=mean(I)
[counts,N]=imhist(I);
counts(1)=0;
i=1;
mu=cumsum(counts);
T(i)=(sum(N.*counts))/mu(end);
T(i)=round(T(i));

% STEP 2: computing Mean above T (MAT) and Mean below T (MBT) using T from
mu2=cumsum(counts(1:T(i)));
MBT=sum(N(1:T(i)).*counts(1:T(i)))/mu2(end);

mu3=cumsum(counts(T(i):end));
MAT=sum(N(T(i):end).*counts(T(i):end))/mu3(end);
i=i+1;
T(i)=round((MAT+MBT)/2);

% STEP 3 to n: repeating step 2 if T(i)~=T(i-1)

```

```

while abs(T(i)-T(i-1))>=1
    mu2=cumsum(counts(1:T(i)));
    MBT=sum(N(1:T(i)).*counts(1:T(i)))/mu2(end);

    mu3=cumsum(counts(T(i):end));
    MAT=sum(N(T(i):end).*counts(T(i):end))/mu3(end);

    i=i+1;
    T(i)=round((MAT+MBT)/2);
    Threshold=T(i);
end

% Normalizing the threshold to the range [i, 1].
level = (Threshold - 1) / (N(end) - 1);
MAT = MAT / (N(end) - 1);
MBT = MBT / (N(end) - 1);
End

```

Function to find the eigenvalues of hessian matrix:

```

function [lamdaplus,lamdaminus]=lamdafind(gxx,gyy,gxy)
% This function perform finding eigen values of hessian matrix and output
% maximum and minimum eigen value as lamdaplus and lambaminus
% gxx1: Second derivative over x
% gyy1: Second derivative over y
% gxy : derivative over x and y
H=[gxx gxy;gxy gyy];

%Obtain eigen values
lamda=eig(H);
%Obtain maximum and minimum lamda values
if lamda(1)>lamda(2)
    lamdaplus = lamda(1);
    lamdaminus = lamda(2);
else if lamda(1)<lamda(2)
    lamdaplus = lamda(2);
    lamdaminus = lamda(1);
else
    lamdaplus = lamda(1);
    lamdaminus = lamda(2);
end
end
end

```